FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – III
PAPER –1
TEST DATE: 01-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –03, 18 – 20, 35 – 37):* This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(04 – 07, 21 – 24, 38 – 41)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(08 – 11, 25 – 28, 42 – 45):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(12 – 17, 29 – 34, 46 – 51):* This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

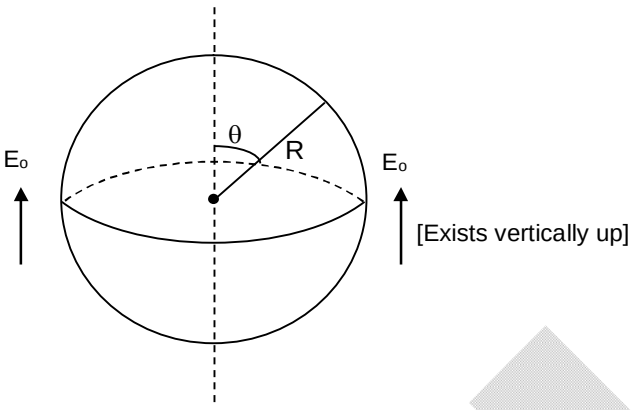
Physics

PART – I

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A neutral conducting solid sphere of radius R is kept in a uniform electric field E_o as shown. Then



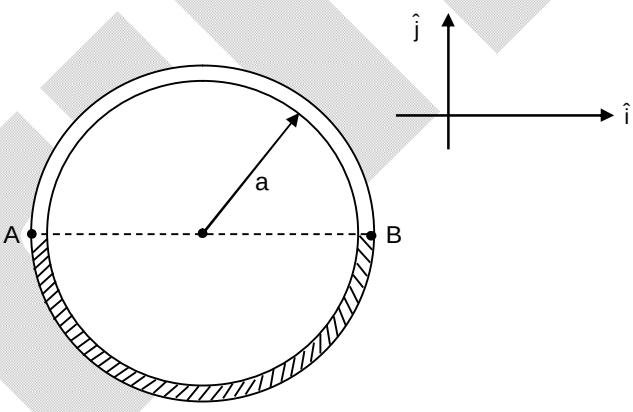
(A) Electric field just outside the surface as a function of θ will be $E = E_0 \cos\theta$

(B) Electric field just outside the surface as a function of θ will be $E = 3E_0 \cos\theta$

(C) Dipole moment of the polarized sphere will be $P = 4\pi \in_0 E_0 R^3$

(D) Electric field due to induced electric charges inside the sphere is uniform

2. Figure shows a conducting neutral ring of radius 'a' made from two different wires of uniform cross-section as shown. A magnetic field exists only inside the ring which varies with time as $\vec{B} = kt\left(-\hat{k}\right)$. Neglect any charge that appears on the surface of the ring. The upper half of the ring has resistance twice as compared to the lower half. The answer the following.



(A) for a uniform ring (same resistivity throughout) potential difference between A and B is zero.

(B) for a uniform ring (same resistivity throughout) potential of B is higher than potential of A.

(C) for the given case (as in diagram), conservative electric field exists along the tangent of the circular ring.

(D) for the given case (as in diagram), potential difference between B and A, $V_{BA} = \dfrac{\pi k a^2}{6}$.

3. A charged particle $\left(q = 2\times10^{-6}C, m = 10^{-9}kg\right)$ is released from origin under uniform magnetic field $\left(\vec{B} = 5mT\,\hat{k}\right)$ and electric field $\left(\vec{E} = 5N/C\,\hat{j}\right)$ as shown. Then during subsequent motion.



(A) Maximum speed the charged particle can achieve will be 10^3 m/s.
(B) Maximum speed the charged particle can achieve will be 2×10^3 m/s.
(C) When it achieves maximum displacement along Y-axis its angular momentum about origin will be zero.
(D) When it achieves maximum displacement along Y-axis its angular momentum about origin will be $4\times10^{-4}kg\,m/s\left(-\hat{k}\right)$

SECTION – A
(One Options Correct Type)

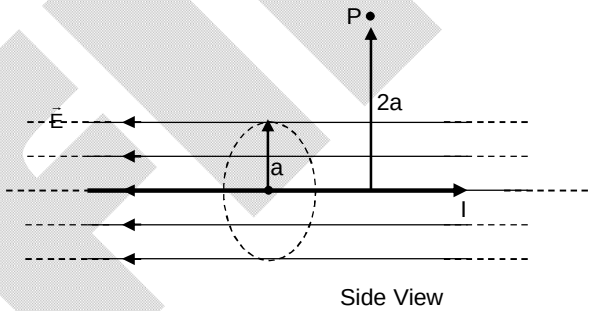
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

4. The radius of the orbit of an electron in hydrogen like atom is $4a_o$ where a_o is the Bohrs radius and its angular momentum is $\frac{2h}{\pi}$. (h= plank's constant R = Rydberg constant). The wavelength this atom would not be emitting as it de-excites itself is

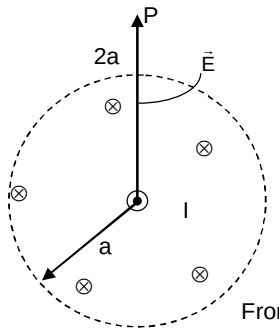
(A) $\frac{9}{7R}$ (B) $\frac{1}{15R}$

(C) $\frac{9}{24R}$ (D) $\frac{9}{128R}$

5. Figure shows a thin long conductor carrying current I from left to right as shown, electric field is also set up in a cylindrical region of radius a surrounding the conductor and runs from right to left as shown. Electric field is varying only with time as E = Kt, where K is a constant P is a point at perpendicular distance of 2a from the wire. If magnetic field at P is found to be zero, value of K is



Side View



Front View

(A) $K = \frac{I}{4\pi a^2 \in_0}$ (B) $K = \frac{I}{\pi a^2 \in_0}$

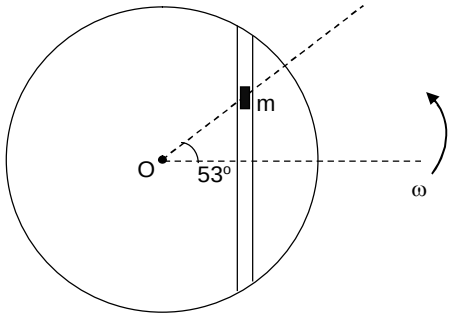
(C) $K = \frac{4I}{\pi a^2 \in_0}$ (D) None of these

6. A tuning fork of frequency 150 Hz is used to find the speed of sound in resonance column apparatus. It resonates in 1st and 2nd harmonic when air columns have length $\ell_1 = (50 \pm 0.5)$cm and $\ell_2 (150 \pm 0.1)$cm. Then find the speed of sound.

(A) $v = (300 \pm 1.8)$m/s (B) $v = (300 \pm 18)$m/s

(C) $v = (300 \pm 9)$m/s (D) $v = (300 \pm 1.2)$m/s

7. Figure shows a horizontal platform having a grove along its chord as shown. A particle which just fits into the grove is placed as shown. Now the platform is given angular acceleration $\alpha = 2$ rad/sec^2 from rest. Co-efficient of friction between the particle and groove is $\mu = 0.8$. Neglect gravity. The particle starts sliding wrt the platform at time 't'. Then



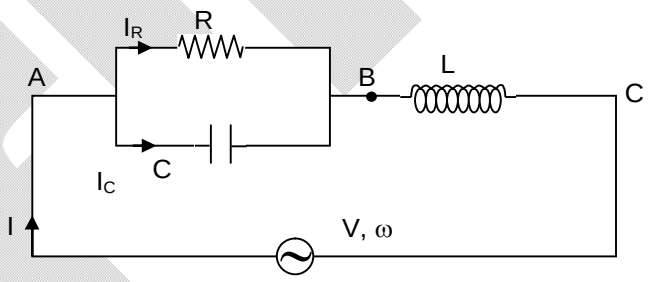
(A) t = 0 (B) $t = \sqrt{\frac{31}{16}}$ sec

(C) $t = \sqrt{\frac{31}{8}}$ (D) $t = \sqrt{\frac{62}{16}}$ sec

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

8. In the given circuit it is given that $V_{AB} = V_{BC} = V_{AC}$ and the line current is I as shown. Mark the correct option.

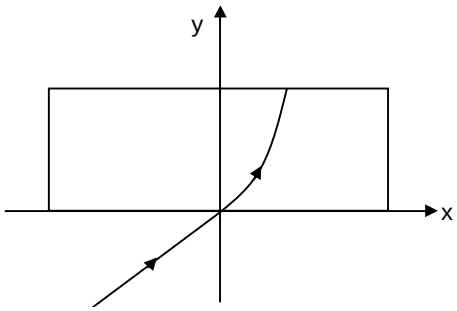


List - I		List - II	
(P)	Phase diff. between V_{AB} and V_{BC}	(1)	π/6
(Q)	Phase diff. between I and I_C	(2)	π/3
(R)	Phase diff between I and V_{AB}	(3)	2π/3
(S)	Phase diff between I_C and V_{BC}	(4)	π/2
		(5)	π/4

Codes:

	P	Q	R	S
(A)	3	2	1	4
(B)	4	2	1	1
(C)	3	2	4	1
(D)	3	2	2	1

9. A slab occupies the region between y = 0 and y = R = 1 m. Rays of light traveling in the XY plane enter the slab at the origin at angle of incidence i and are found to travel in circular path. Consider the following case.
Case 1 : The refractive index of the slab, μ, varies from 1 at y = 0 to infinity at y = R and i = 90.
Case 2 : The refractive index of the slab, μ, varies from infinity at y = 0 to 1 at y = R and i = 0.

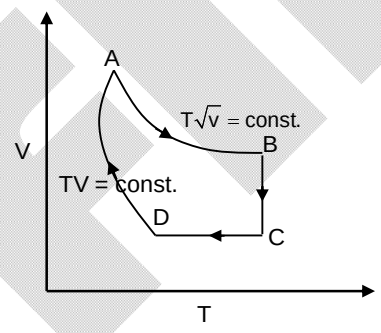


List - I		List - II	
(P)	In case 1, $1/\mu$ at y = R/4	(1)	4/5
(Q)	In case 1, x at y = 2R/5	(2)	1/5
(R)	In case 2, $1/\mu$ at y = R/2	(3)	3/4
(S)	In case 2, x at y = 3R/5	(4)	1/2
		(5)	1/4

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	1	4	3	2
(C)	2	2	1	4
(D)	3	1	4	2

10. In the VT diagram shown, we have 4 processes forming a thermodynamic cycle. The gas is mixture of hydrogen & helium in mole ratio 1 : 1. Comment on the entropy of gas

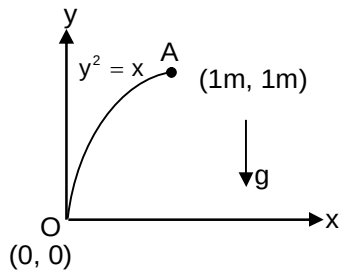


List - I		List - II	
(P)	AB	(1)	Entropy at end is same as entropy at beginning
(Q)	BC	(2)	Entropy at end is less than entropy at beginning
(R)	CD	(3)	Entropy at end is more than entropy at beginning
(S)	DA	(4)	Entropy increases and then decreases
		(5)	Entropy decreases and then increases

Codes:

	P	Q	R	S
(A)	1	2	2	3
(B)	2	2	2	3
(C)	4	5	2	3
(D)	1	2	2	2

11. A rough track connects point O (0m, 0m) to point A(1m, 1m). External force F always tangential to path pulls object of mass 2 kg from point O(0m, 0m) to point A(1m, 1m) very slowly. Coefficient of friction between block and track is 0.5.



List - I		List - II	
(P)	Work done by frictional force on block magnitude wise in Joules	(1)	30
(Q)	Normal reaction force between block and track at $x = \frac{1}{2}$m in Newton	(2)	$20\sqrt{\frac{2}{3}}$
(R)	Work done by external force F on block magnitude wise in Joules	(3)	10
(S)	Net reaction force between block and track at $x = \frac{3}{4}$m in Newton	(4)	$20\sqrt{\frac{5}{13}}$
		(5)	$5\sqrt{15}$

If equation of track is $y^2 = x$ then choose the correct combination from List I & List II.

Codes:

	P	Q	R	S
(A)	3	2	1	5
(B)	4	2	5	1
(C)	1	2	3	4
(D)	1	2	5	3

SECTION – B

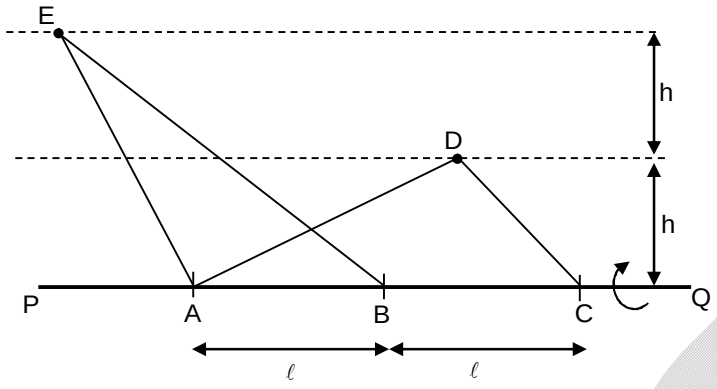
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

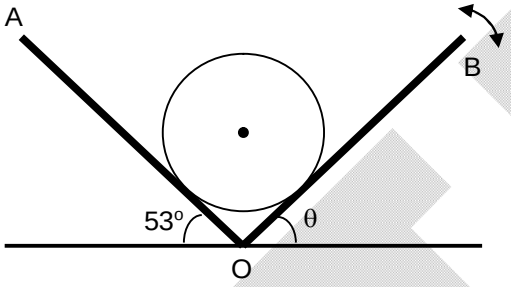
12. In a standard YDSE set up, the separation between 1st and 11th bright fringe on a screen along Y-axis is 2.5 mm. The screen is at a distance of 3 m from the interfering sources. Ratio of intensity at 0.1 mm from central maxima and maximum intensity on the screen is found to be $\cos^2\left(\frac{2\pi}{k}\right)$.

Find the value of k. [Take λ = 5000 $\overset{o}{A}$]

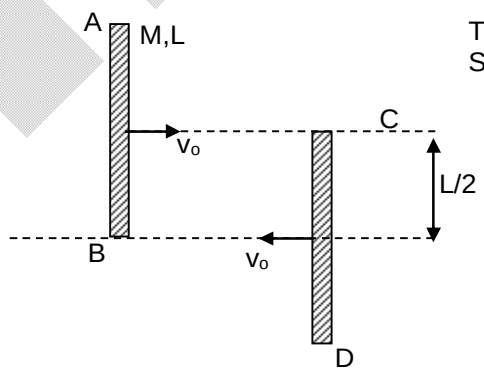
13. Two uniform triangular plates AEB and ACD of same thickness and made of the same material are shown in figure. Heights and length of base are as indicated. If moment of inertia of plate AEB and ACD about the axis PQ (lying in the same plane) are I_1 and I_2 respectively then find the value of I_1 / I_2.



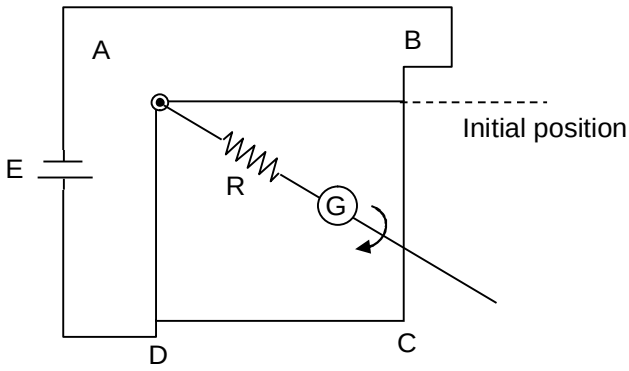
14. In the given figure, a sphere of mass 1 kg is kept at rest with the help of 2 inclined planes. Plane OB can make a variable angle θ from horizontal. If N be the normal contact form plane OB, find the minimum value of N in newton.



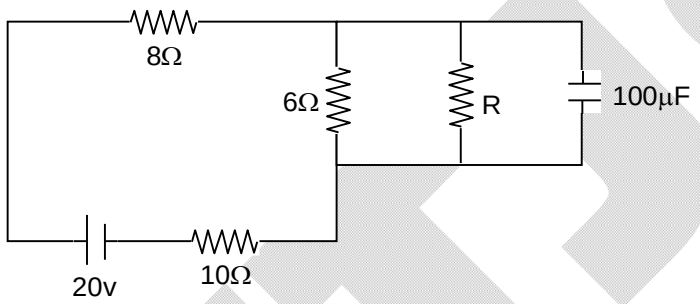
15. Two identical uniform rods are moving towards each other with their lengths perpendicular to their velocities as shown. Half of both rods are involved in collision process as indicated Co-efficient of restitution for the impact is $e = \frac{1}{2}$. Then the velocity of end B of the rod just after impact is found to be $\frac{mv_0}{n}$. Find m + n.



16. ABCD is a square frame Resistances of wires AB, BC, CD and DA are 100 Ω, 400Ω, 500Ω and 200Ω respectively. A straight conducting wire having a galvanometer starts rotating about pivoted point. A form initial position as shown with $\omega = \frac{\pi}{360}$ rad/sec. Other end of the conducting wire (rotating) is always in sliding contact with a side of the square. The time after which the galvanometer shows zero deflection in seconds is:



17. In the circuit shown find the charge in the on the capacitor when power loss through resistor R is maximum at steady state.



Chemistry

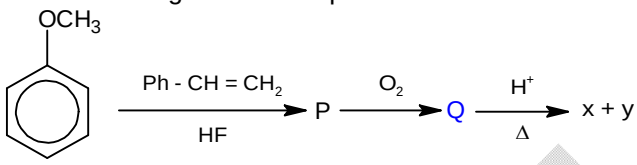
PART – II

SECTION – A
(One or More than one correct type)

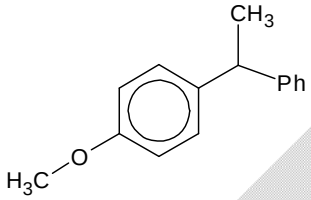
*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

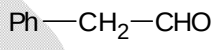
18. Which among the following statement is/are correct?
(A) pH of 10^{-8} M HCl is equal to 8 at 25°C
(B) Conjugate base of $H_2PO_4^-$ is HPO_4^{2-}
(C) pH of $0.1\,M\,NaCl(aq) = \frac{1}{2}pkw$
(D) Ionization of water increases with decrease in temperature

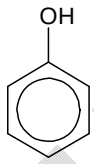
19. In the following reaction sequence

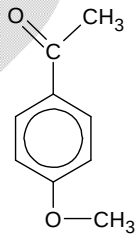


Which of the following is/are not among P, Q, X and Y?

(A) 

(B) 

(C) 

(D) 

20. HI cannot be prepared by action of conc. H_2SO_4 on KI because
(A) HI is stronger acid than H_2SO_4
(B) H_2SO_4 is an oxidizing agent
(C) HI is strong reducing agent
(D) H_2SO_4 is stronger acid than HI

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

21. $I_{2(s)} / I^-_{(aq)}(0.1\,M)$ half cell is connected to a $H^+_{(aq)} / H_2(1bar) / Pt$ half cell and emf is found to be 0.7714V. If $E^0_{I_2/I^-} = 0.535V$ then find pH of H^+ / H_2 half cell.
(A) 1
(B) 3
(C) 5
(D) 7

22. $A_{(aq)} + Zn \longrightarrow B_{(g)}$

$A_{(aq)} + C \xrightarrow{\Delta} PH_3$

$A_{(aq)} + NH_4Cl \longrightarrow D_{(g)}$

Find compound A if it import yellow flame colour.

(A) LiOH (B) KOH

(C) NaOH (D) RbOH

23. The wave function of orbital of a H-like atom is

$\psi_{2S} = \frac{1}{4\sqrt{2\pi}} Z^{3/2} (2 - zr) e^{-zr/2}$ (r is in A^o)

Which of the following will be radius of nodal surface for He^+ ion?

(A) 1 A^o (B) 2A^o

(C) 2.5 A^o (D) 4 A^o

24. Which of the following is not true about diborane?

(A) It has two bridging hydrogen and four terminal hydrogens

(B) When methylated the product is $Me_4B_2H_2$

(C) The bridging hydrogen are in a plane perpendicular to rest B – H bonds

(D) All six B – H bond lengths are equal.

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the List-I with correct properties of List-II.

List - I		List - II	
(P)	0.1 M $Al_2(SO_4)_3$	(1)	Solution with highest boiling point
(Q)	0.1 M $AlPO_4$	(2)	Van't Hoff factor if greater than 1
(R)	0.1 M Urea	(3)	Solution with lowest osmotic pressure
(S)	0.1 M $MgCl_2$	(4)	Solution with lowest freezing point
		(5)	Solution with lowest boiling point

Codes:

	P	Q	R	S
(A)	1,4	2	5	2
(B)	1,2,4	2	3,5	2
(C)	4	2	3,5	4
(D)	2,4,5	3,4	2,5	4

26. If P_n = potential energy, E_n = total Energy F_n = frequency, Z = atomic number, V_n = velocity in n^{th} orbit, T_n = time period in n^{th} orbit, then match the List – I to List-II.

List - I		List - II	
(P)	$E_n \propto r^y, y = ?$	(1)	$\frac{1}{2}$
(Q)	E_n / P_n	(2)	1
(R)	$\frac{1}{F_n - x} \propto z, x = ?$	(3)	2
(S)	$(V_n \times T_n)^t \propto r, t = ?$	(4)	-1
		(5)	0

Codes:

	P	Q	R	S
(A)	4	1	1	2
(B)	1	5	2	1
(C)	1	1	2	1
(D)	2	2	5	4

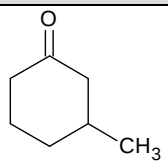
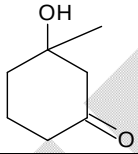
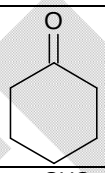
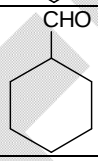
27. Match the following:

List - I (Elements with electron in K, L, M, N shells)		List - II (Statements)	
(P)	W (2, 8, 7)	(1)	Paramagnetic
(Q)	X (2, 8, 18, 8)	(2)	3rd group element
(R)	Y (2, 8, 14, 2)	(3)	Last electron does not enter in outermost shell
(S)	Z (2, 8, 18, 25, 8, 2)	(4)	Reactive non metal
		(5)	Diamagnetic

Codes:

	P	Q	R	S
(A)	2,3	4	4,5	1,2,5
(B)	3,4	1	2,1	1,2,3,1
(C)	1	2	3	4
(D)	1,4	5	1,3	1,2,3

28. Match List-I with correct properties of List-II:

List - I		List - II	
(P)		(1)	Forms hydrocarbon when reacts with NH_2NH_2 / KOH
(Q)		(2)	Forms diol when reduced with $LiAlH_4$
(R)		(3)	Forms monohydric alcohol when treated with $NaBH_4$
(S)		(4)	Forms Unsaturated ketone when heated on Cu at 300^oC
		(5)	Forms toluene with Zn

Codes:

	P	Q	R	S
(A)	2,1	1	1,4	5
(B)	1,3	1	2,4	1,4
(C)	1,3	2,4	1,3	1,3
(D)	2	1	4	5

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

29. Among the following the total number of species which are paramagnetic:
$O_2, S_2, CN^-, NO^+, O_2^{++}, B_2, N_2^{--}, O_2^+, NO, NO_2, N_2O$

30. ΔH_f^0 HF(g) $= -275\,kJ/mol$
ΔH_f^0 $H^+(aq) = 0$
ΔH_f^0 $F^-(aq) = -335\,kJ/mol$
Amount of energy released in kJ when 1 gm of HF(g) is converted into H^+ (aq) and F^- (aq) is -x kJ. Then find the value of x.

31. Among $NH_3, PH_3, AsH_3, SbH_3, H_2O, H_2S, H_2Se, H_2T_e$ and CH_4. The total number of compounds in which H – M – H bond angle is > 100°.

32. A monohydric alcohol has two chiral carbons, if all the carbons are sp^3 hybridized, find minimum number of carbon atoms possible in alcohol.

33. A non-cyclic aldehyde 'A' having molecular formula C_xH_yO has one chiral carbon. 'A' does not participate in aldol condensation. The minimum value of 'x' is:

34. Quantitative estimation of ozone is carried out by passing ozonized oxygen in a KI solution buffered with borax (pH = 9.6) and I_2 liberated out is measured using sodium thiosulphate. 10 litres of a sample of ozonized oxygen (measured at S.T.P.) when passed through KI solution, the I_2 liberated requires 2 litres of $\frac{M}{5.6}Na_2S_2O_3$ solution. Volume of ozone in litres in ozonized oxygen is: (consider at STP, 1 mole gas = 22.4 litre)

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

35. Which of the following statement(s) always hold(s) good?

(A) $P\left(\overline{A}\overline{B}\right)-P\left(\overline{A\cup B}\right)$ denotes the probability that exactly one of the two events A and B occurs.

(B) For any two events A and B, $P(A)+P\left(\overline{A}B\right)+P\left(\overline{A\cup B}\right)=1$.

(C) Two mutually exclusive events A and B for which $P(A)>0$ and $P(B)>0$ can be independent.

(D) If $P\left(B/\overline{A}\right)=P(B/A)$ then A and B are independent.

36. The determinant $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$ is equal to zero, if:

(A) a, b, c are in AP
(B) a, b, c are in GP
(C) α is a root of the equation $ax^2+bx+c=0$
(D) $(x-\alpha)$ is a factor of $ax^2+2bx+c$

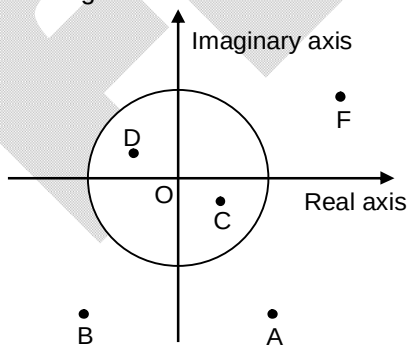
37. Let $f(x)=\begin{cases} x\left[\frac{1}{x}\right]+x[x] & \text{if } x\neq 0 \\ 0 & \text{if } x=0 \end{cases}$, where [x] denotes the greatest integer function, then the correct statements are:

(A) limit exists for $x=-1$.
(B) f(x) has a removable discontinuity at x = 1
(C) f(x) has a non removable discontinuity at x = 2.
(D) f(x) is discontinuous at all positive integers

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

38. The diagram shows several numbers in the complex plane. The circle is the unit circle centered at the origin. One of these numbers is the reciprocal of F, which is:



(A) A
(B) B
(C) C
(D) D

39. There are m points on a straight line AB and n points on the line AC, none of them being the point A. Triangles are formed with these points as vertices, when
(i) A is excluded (ii) A is included
The ratio of number of triangles in the two cases is:

(A) $\frac{m+n-2}{m+n}$ (B) $\frac{m+n-2}{m+n-1}$

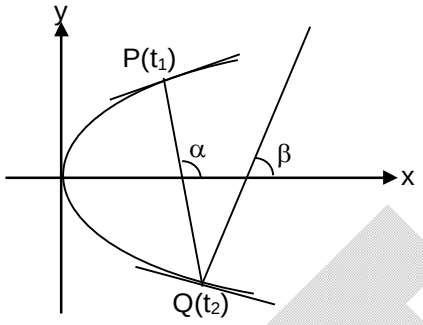
(C) $\frac{m+n-2}{m+n+2}$ (D) $\frac{m(n-1)}{(m+1)(n+1)}$

40. Let $e^{f(x)} = \ln x$. If g(x) is the inverse function of f(x) then g'(x) equals to:

(A) e^x (B) $e^x + x$

(C) $e^{(x+e^x)}$ (D) $e^{(x+\ln x)}$

41. If the normal to a parabola $y^2 = 4ax$ at P meets the curve again in Q and if PQ and the normal at Q makes angles α and β respectively with the x-axis, then $\alpha(\tan\alpha + \tan\beta)$ has the value of equal to:



(A) 0 (B) – 2

(C) $-\frac{1}{2}$ (D) – 1

SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

42. Consider the following four pairs of lines in List-I and match them with one or more entries in List-II.

List - I		List - II	
(P)	$L_1 : x = 1+t, y = t, z = 2-5t$ $L_2 : \vec{r} = (2,1,-3) + \lambda(2,2,-10)$	(1)	Non-coplanar lines
(Q)	$L_1 = \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(2)	Lines lie in a unique plane
(R)	$L_1 : x = -6t, y = 1+9t, z = -3t$ $L_2 : x = 1+2s, y = 4-3s, z = s$	(3)	Infinite planes containing both the lines

(S)	$L_1 = \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 = \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(4)	Lines do not intersect
		(5)	Lines are coincident

Codes:

	P	Q	R	S
(A)	4,5	2,3	1	4
(B)	3,5	2	2,4	1,4
(C)	3	3	4	1
(D)	3,1	2	2,3	1,5

43. Match the List-I with List-II.

List - I		List - II	
(P)	Let $f(x) = \int x^{\sin x}(1 + x\cos x \cdot \ln x + \sin x)dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$ then the value of $f(\pi)$ is	(1)	Rational
(Q)	Let $g(x) = \int \frac{1+2\cos x}{(\cos x + 2)^2}dx$ and $g(0) = 0$ then the value of $g\left(\frac{\pi}{2}\right)$ is	(2)	Irrational
(R)	If real numbers x and y satisfy $(x+5)^2 + (y-12)^2 = (14)^2$ then the minimum value of $\sqrt{(x^2 + y^2)}$ is	(3)	Integral
(S)	Let $k(x) = \int \frac{(x^2+1)dx}{\sqrt[3]{x^3 + 3x + 6}}$ and $k(-1) = \frac{1}{\sqrt[3]{2}}$ then the value of $k(-2)$ is	(4)	Prime
		(5)	Composite

Codes:

	P	Q	R	S
(A)	1,3	1,4	2	3
(B)	3,4	4	5	1,3
(C)	1,5	2	1	1,3
(D)	2	1	1,3	1,3,4

44. Match the following:

List - I		List - II	
(P)	Number of irrational terms in the expansion of $\left(\sqrt{2} + \sqrt{3}\right)^{15}$ is equal to	(1)	6
(Q)	If $\sum_{r=1}^{n} {}^nC_r 3^r$ is equal to 4095 then n equals	(2)	7
(R)	If ${}^nP_r = \left({}^{n-1}C_{r-1} + {}^{n-1}C_r\right) \cdot 5040$ then the value of r equals	(3)	15
(S)	Number of five digit integers such that the product of their digits is 2000 is 2 K then K equals	(4)	16
		(5)	17

Codes:

	P	Q	R	S
(A)	4	1	2	3
(B)	2	2	2	5
(C)	4	1	3	5
(D)	1	2	3	4

45. Let $\lim\limits_{T\to\infty}\frac{1}{T}\int\limits_0^T(\sin x+\sin ax)^2\,dx=L$ then:

List - I		List - II	
(P)	For a = 0, the value of L is	(1)	0
(Q)	For a = 1 the value of L is	(2)	1/2
(R)	For $a=-1$ the value of L is	(3)	1
(S)	$\forall a\in R-\{-1,0,1\}$ the value of L is	(4)	2
		(5)	3/2

Codes:

	P	Q	R	S
(A)	2	4	1	3
(B)	4	2	1	5
(C)	1	2	3	4
(D)	5	1	4	3

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

46. Let $\vec{a}=3\hat{i}+2\hat{j}+4\hat{k};\vec{b}-=2(\hat{i}+\hat{k})$ and $\vec{c}=4\hat{i}+2\hat{j}+3\hat{k}$. If the equation $x\vec{a}+y\vec{b}+z\vec{c}=\alpha(a\hat{i}+y\hat{j}+z\hat{k})$ has a non-trivial solution, then find the sum of all distinct, possible values of α.

47. If the value of the definite integral $\int\limits_{-1}^{1}\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\cdot\left(\cot^{-1}\frac{x}{\sqrt{1-(x^2)^{|x|}}}\right)dx=\frac{\pi^2(\sqrt{a}-\sqrt{b})}{\sqrt{c}}$

where $a,b,c\in N$ in their lowest form, then find the value of (a + b + c).

48. Let the equation $(a-1)x^2=x(2b+3)$ be satisfied by three distinct values of x, where $a,b\in R$. If $f(x)=(a-1)x^3+2(2b+3)x^2+2x+1$, and $f(g(x))=6x-7$ where g(x) is a linear function, then find the value of $g'(2012)$.

49. Let $f:R\to\left(0,\frac{2\pi}{3}\right]$ defined as $f(x)=\cot^{-1}(x^2-4x+\alpha)$. Find the smallest integral value of α such that f(x) is into function.

50. For $m,n\in N$, let the 6th term in the expansion of the binomial $\left(\sqrt{2^{\log(10-3^n)}}+\sqrt[5]{2^{(n-2)\log 3}}\right)^m$ is equal to 21, if binomial coefficient of 2nd, 3rd and 4th term in the expansion represent respectively the 1st, 3rd and 5th terms of an arithmetic progression. Find (m +n). (Assume that the symbol log stands for logarithm to the base 10.)

51. Let $V_{n+1}=3V_n-2V_{n-1}\ \forall n\in N$ and $V_0=2, V_1=3$, then value of $\left(\frac{V_6}{13}\right)$ is: